

Tasks 1A–1C: Exploratory Data Analysis of Wine Quality

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Overview

This document presents the exploratory data analysis (EDA) conducted as part of 2034 in the wine quality prediction project. The goal is to investigate how properties — including alcohol and residual sugar — of wines relate to quality in both red and white wines. This task analysis will guide future predictive modelling.

Task 1A: Initial Data Analysis

Quality Distribution – Red Wine

The first analysis focused on the distribution of the `quality` variable in the red wine dataset. The values range from 3 to 8, with most samples concentrated around 5 and 6.



`../figures/red_wine_quality.png`

Figure 1: Distribution of Quality Scores in Red Wine Dataset

Observation: The majority of red wine samples are rated 5 or 6 in quality. Higher scores (7 and 8) occur less frequently, and very few samples are rated below 4.

Quality Distribution – White Wine

The same distribution analysis was applied to the white wine dataset.



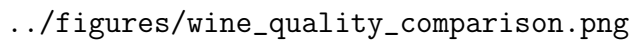
`../figures/white_wine_quality.png`

Figure 2: Distribution of Quality Scores in White Wine Dataset

Observation: Like red wine, white wine samples mostly fall in the 5–6 quality range. However, white wine has a wider spread, with a noticeable number of samples rated 7 or 8, and more rated 4 as well.

Red vs White Comparison

To compare the two datasets, the quality distributions were plotted side-by-side.



../figures/wine_quality_comparison.png

Figure 3: Side-by-Side Distribution of Red and White Wine Quality

Interpretation:

- Red wine quality scores are tightly clustered around 5 and 6.
- White wine exhibits more diversity, with higher and lower quality ratings more common.
- This variation may be influenced by grape types, fermentation techniques, or sample size (white wine has over three times more records).

Task 1B: Alcohol Content and Wine Quality

Overview

This task examines whether alcohol content is associated with perceived quality. Alcohol levels were categorised as `low`, `mid`, or `high` based on the dataset's mean and standard deviation:

- `low`: below ($\text{mean} - \text{standard deviation}$)
- `mid`: within one standard deviation of the mean
- `high`: above ($\text{mean} + \text{standard deviation}$)

The new variable `alcohol_cat` enables comparison of quality across these groups.

Red Wine



Figure 4: Red Wine Quality by Alcohol Category

Observation:

- Red wines with higher alcohol levels tend to have higher quality scores.

- There is a clear upward shift in the distribution of quality from low to high alcohol groups.
- Low-alcohol red wines are rarely rated above 6.

White Wine

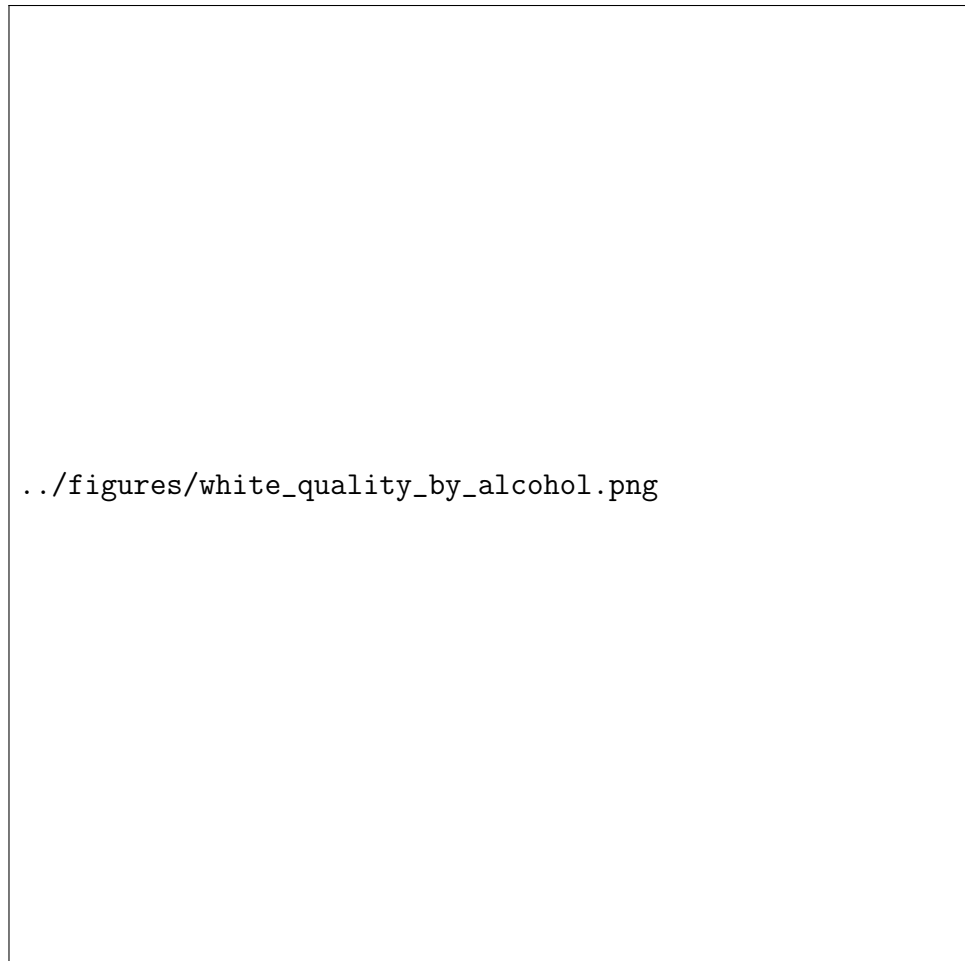


Figure 5: White Wine Quality by Alcohol Category

Observation:

- A similar, though less pronounced, trend is observed in white wine.
- Higher alcohol white wines are rated slightly better, on average.
- The distribution is narrower than in red wines.

Interpretation

- Alcohol content is positively associated with wine quality for both types.
- This may be due to alcohol signalling better fermentation or grape maturity.
- `alcohol` appears to be a strong candidate for predictive modelling.

Task 1C: Residual Sugar and Sweetness Classification

Overview

This task explores whether residual sugar affects perceived quality. A new binary feature `isSweet` was created by splitting each dataset at the median residual sugar level:

- `isSweet` = 1 if residual sugar $\geq$ median
- `isSweet` = 0 otherwise

This provides a roughly balanced classification of wines into “dry” and “sweet”.

Red Wine



Figure 6: Red Wine Quality by Sweetness (`isSweet`)

Observation:

- Sweet red wines (`isSweet`=1) score slightly lower on average.
- Quality distributions for dry and sweet red wines overlap substantially.

White Wine



Figure 7: White Wine Quality by Sweetness (`isSweet`)

Observation:

- White wine shows a clearer quality distinction.
- Dry wines tend to have higher scores than sweet ones.
- Sweet white wines are more frequently rated 5 or lower.

Interpretation

- Residual sugar has a weak negative correlation with quality.
- This suggests that drier wines may be perceived as higher quality, possibly due to taste expectations.
- `isSweet` could be a useful feature in predictive models, though its influence is less than alcohol.

Task 1D: Quality vs Sweetness

This task evaluates whether sweet wines are perceived as higher or lower quality, using the binary `isSweet` feature created in Task 1C.

Red Wine: The distributions for dry and sweet red wines are very similar. Sweet red wines show a slightly lower median quality score, but the overall shift is minimal.

White Wine: White wines show a clearer separation between dry and sweet. Dry white wines tend to score higher, while sweet white wines are more concentrated in the lower range (mostly 5 or below).

Conclusion: Sweetness appears to ****negatively impact**** perceived quality for white wines, but ****has minimal effect**** in reds. The `isSweet` feature may therefore contribute more to prediction in white wine modelling.

Task 2: Correlation Analysis

To identify which variables might be most useful for predicting wine quality, correlation matrices were produced for both red and white wine datasets. The Pearson correlation coefficient was used as it measures **linear relationships** between continuous features. This is appropriate for the physicochemical variables provided.

Red Wine Correlation Matrix

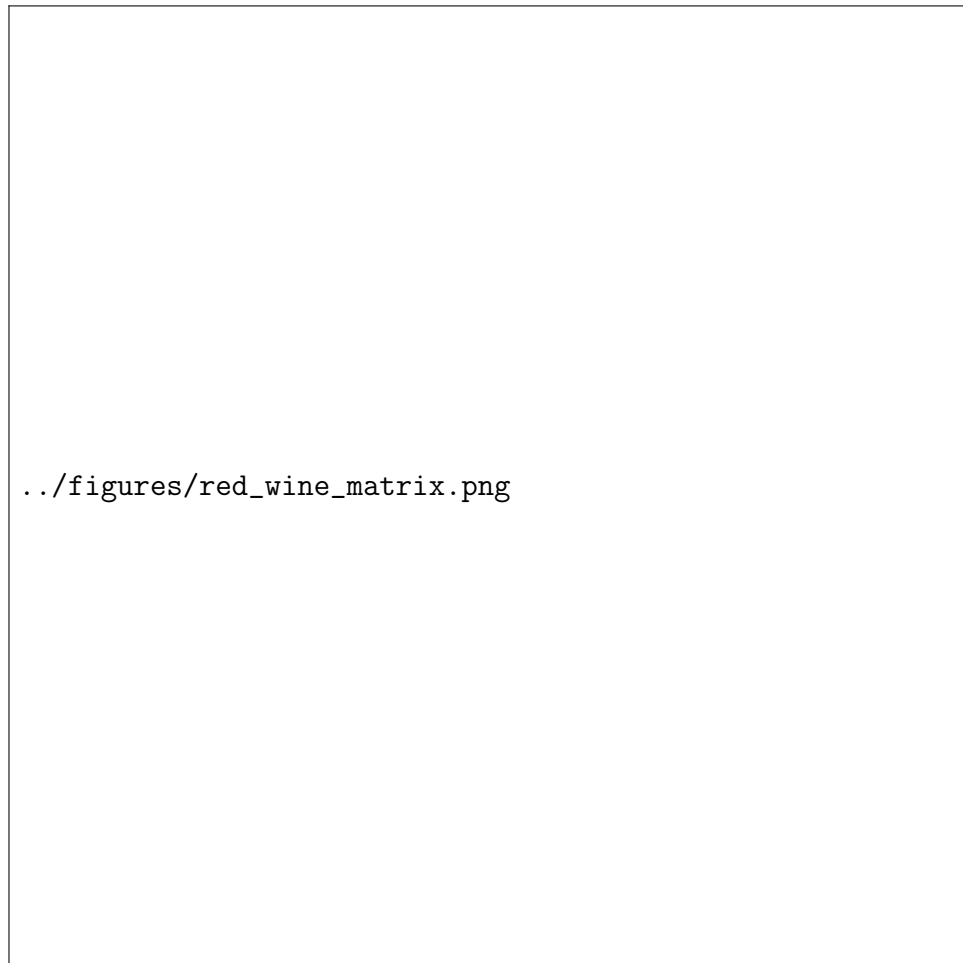


Figure 8: Correlation Matrix for Red Wine (Pearson)

Key Findings:

- **Alcohol** shows the strongest positive correlation with quality ($r \approx 0.48$).
- **Volatile acidity** is moderately negatively correlated with quality.
- **Sulphates** and **citric acid** show weak-to-moderate positive correlations.
- **Free sulphur dioxide** and **total sulphur dioxide** are highly correlated with each other (redundant).

White Wine Correlation Matrix



Figure 9: Correlation Matrix for White Wine (Pearson)

Key Findings:

- **Alcohol** again has the strongest positive correlation with quality ($r \approx 0.44$).
- **Residual sugar** and **density** are weakly negatively correlated with quality.
- Sulphur dioxide variables remain strongly correlated.

Interpretation

- **Alcohol** is a consistently strong feature and will likely be a key predictor in modelling.
- **Volatile acidity** and **density** may serve as negative predictors.
- Highly correlated variable pairs (e.g., **free_S02** and **total_S02**) may be reduced to avoid redundancy.
- This correlation analysis will inform feature selection for the machine learning models in Task 3.

Task 3 – Predictive Modelling

The goal of this task was to experiment with machine learning approaches to predict wine quality based on physicochemical variables.

Modelling Approaches

Two main types of models were explored:

- **Binary Classification:** Predict whether a wine is “high quality” using thresholds on the quality score (e.g., ≥ 6).
- **Regression:** Predict the exact quality score as a continuous numerical value.

Thresholds for Classification

To explore the impact of class definition, three thresholds were tested:

- **Threshold ≥ 5 :** Too lenient — the dataset becomes highly imbalanced.
- **Threshold ≥ 6 :** Balanced and realistic — this was the best-performing split overall.
- **Threshold ≥ 7 :** More strict — harder task due to fewer high-quality samples.

Each model was trained using a Random Forest, which was chosen for its robustness, non-linear capabilities, and ability to rank feature importance.

Feature Processing

All features were numeric and standardised using `StandardScaler` from scikit-learn. The original `quality` column was excluded from the input features.

Task 4 – Model Evaluation

Each model was evaluated using **5-fold cross-validation** and tested on a held-out 20% split. Evaluation metrics varied by model type:

Binary Classification

Threshold ≥ 5 :

- F1-score (CV): 0.979
- Accuracy: 95%
- ROC AUC: 0.910
- *Note: Model simply predicts class 1 — poor for real classification*

Threshold ≥ 6 (Best Result):

- F1-score (CV): 0.817
- Accuracy: 80%
- ROC AUC: 0.902
- Balanced performance across both classes

Threshold ≥ 7 :

- F1-score (CV): 0.510
- Accuracy: 94%
- ROC AUC: 0.955
- High precision, low recall on minority class (class imbalance problem)

Regression Evaluation

The model predicted the actual wine quality as a continuous value. Results:

- RMSE (CV): 0.609
- RMSE (Test): 0.548
- MAE: 0.422
- R^2 : 0.540

Conclusion

- Classification using threshold ≥ 6 yielded the most reliable performance, with balanced precision and recall and a high AUC.
- Regression worked reasonably well but was limited by the narrow range and subjectivity of quality scores.
- Future work could include hyperparameter tuning, model stacking, or testing other classifiers like Gradient Boosting or SVM.